

THIRD SEMESTER**B.Tech. (ECE)**

END SEM. EXAMINATION, Nov.-Dec. 2024

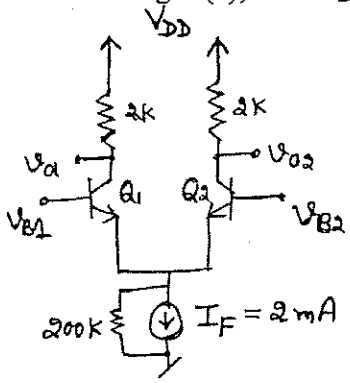
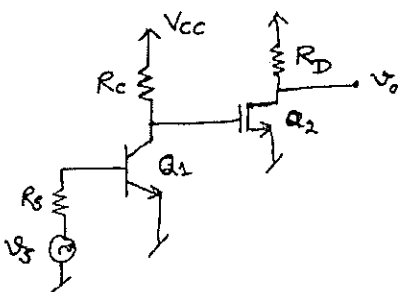
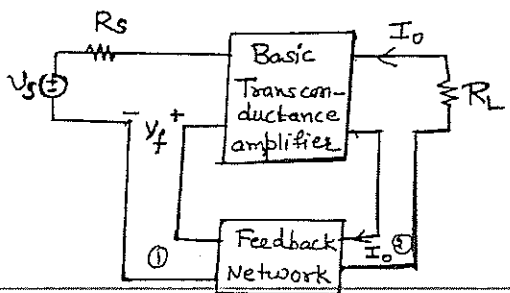
Course Code: ECECC07

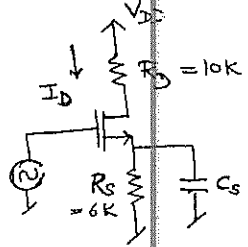
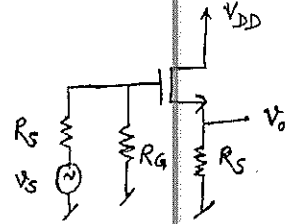
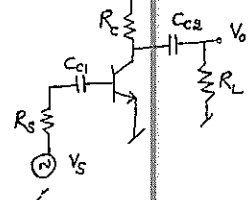
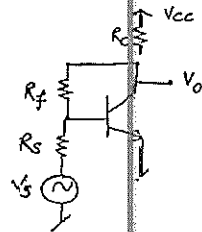
Course Title: Microelectronics

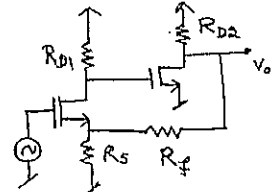
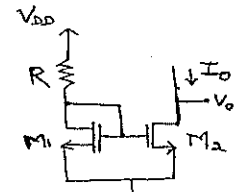
Time: 03:00 Hrs.

Max. Marks: 40

Note: Attempt all the five questions. All questions carry equal marks. Missing data/information if any will be suitably assumed and mentioned in the answer.

Q. No.	Question	Marks	CO
Q1	Attempt any 2 parts of the following		
1(a)	i) For the Differential Amplifier shown in Fig. 1(a), find A_D and A_{CM}.  Fig. 1(a) ii) Define CMRR for differential amplifier and calculate the same for the circuit shown in Fig. 1(a).	(3+1)	CO1
1(b)	For the circuit shown in Fig. 1(b): i) Draw small signal equivalent circuit. ii) Find overall gain.  Fig. 1(b)	(1+3)	CO1
1(c)	i) A series-series feedback circuit representable by Fig. 1(c) and using ideal trans-conductance amplifier operates with $V_S = 100\text{mV}$, $V_f = 95\text{mV}$ and $I_O = 10\text{mA}$. What are the corresponding values of A and β. Include the unit in each answer.  Fig. 1(c) ii) Find R_{if} in terms of R_i, A and β for series-series topology.	(3+1)	CO3

Q2	Attempt any 2 parts of the following		
2(a)	For the amplifier shown in Fig. 2(a), is biased to operate at $I_D = 1\text{mA}$, $g_m = 1\text{mA/V}$, $R_D = 10\text{k}\Omega$, $R_S = 6\text{k}\Omega$. Neglecting r_o , find the values of C_S that places f_L at 100Hz .	(4)	CO2
	 Fig. 2(a)		
2(b)	For the circuit shown in Fig. 2(b) draw small signal high frequency equivalent circuit and find higher cut of frequency (f_H) by assuming the dominant pole approximation is valid.	(4)	CO2
	 Fig. 2(b)		
2(c)	For the common emitter amplifier shown in Fig. 2(c), find lower cut of frequency (f_L).	(4)	CO2
	 Fig. 2(c)		
Q3	Attempt any 2 parts of the following		
3(a)	Draw the feedback amplifier in signal flow graph form/ block diagram and find gain with feedback (A_f), assuming that negative feedback is applied. Draw the series-shunt topology and find its R_{if} and R_{of} in terms of A and β .	(4)	CO3
3(b)	For the circuit shown in Fig. 3(b), with following parameters: $R_C = 4\text{k}\Omega$, $R_F = 40\text{k}\Omega$, $R_S = 10\text{k}\Omega$, $r_x = 1.1\text{k}\Omega$, transistor current gain (h_{fe}) = 50, find A_f , R_{if} , and R_{of} .	(4)	CO3
	 Fig. 3(b)		

3(c)	For the series-shunt amplifier shown in Fig. 3(c), find A_f , R_{if} , and R_{of} .	(4)	CO3
	 Fig. 3(c)		
Q4	Attempt any 2 parts of the following		
4(a)	Draw the circuit of MOSFET based cascode current mirror and derive the expressions for its output resistance. Also list ideal characteristics of a current mirror.	(4)	CO4
4(b)	i) Draw the circuit of class B push-pull amplifier and derive its relation of conversion (overall) efficiency. ii) Class B push-pull amplifier with $V_{CC} = 25\text{V}$ drives a 8Ω load. Determine the maximum input power, output power, efficiency and power dissipated by each transistor.	(2+2)	CO4
4(c)	Design a MOSFET based current mirror to obtain an output current of $50\mu\text{A}$ (Fig. 4(c)). $V_{DD} = 1.8\text{V}$, $I_{ref} = 50\mu\text{A}$, find R , if M_1 and M_2 are matched with $L = 0.5\mu\text{m}$, $W = 5\mu\text{m}$, $V_{th} = 0.5\text{V}$, $K_n = 250\mu\text{A/V}^2$.	(4)	CO4
	 Fig. 4(c)		
Q5	Attempt any 2 parts of the following		
5(a)	State the Barkhausen Criteria. Draw MOSFET based Wein-bridge oscillator and derive its expressions for condition of sustained oscillation and frequency of oscillation.	(5)	CO5
5(b)	Draw transistor level circuit of a two stage CMOS Op-amp and explain Miller compensation.	(5)	CO5
5(c)	Briefly explain following parameters for an Op-amp: i) Slew rate ii) CMRR iii) PSRR iv) ICMR	(5)	CO5